JOURNAL OF VERBAL LEARNING AND VERBAL BEHAVIOR 12, 51–63 (1973)

# The Reconstructive Process in Sentence Memory[1]

CARLTON T. JAMES, JACK G. THOMPSON
AND
JACQUELINE M. BALDWIN

*Rutgers, The State University, New Brunswick, New Jersey 08903*

This report examines the role of reconstructive processes in the recall of simple sentences. As both recall and normal speech production are constructive activities, any characteristic patterns identified in speech production should influence the constructions produced in recall. Two characteristic patterns were specified: a preference for active over passive sentences, and a tendency to begin a sentence with the most salient noun contained in the semantic situation being communicated. Evidence was obtained for the operation of both biases. Recent sentence-memory hypotheses concentrate on the form in which information is stored, and none of these is consistent with the obtained results. In order to evaluate alternative storage hypotheses, the memorial processes that operate on the memory structure must be specified.

Fillenbaum (1970) has discussed the problem of using memorial techniques to study the form in which sentences are stored in memory. He makes the point that "the distribution of confusions or errors on a memorial task cannot be considered as directly revealing of the structures underlying the strings memorized, or of the similarities between such structures [p. 237]." However, as Fillenbaum notes, the role of memorial processes has been generally ignored, and results have been interpreted as rather directly reflecting the contents of memory. We would like to illustrate the importance of memorial processes by reporting recall data that seem to support none of the recent hypotheses of sentence storage, unless elaborated with some assumptions about reconstruction.

Recent hypotheses about sentence storage have generally assumed that the sentence-as-presented (the surface structure) is not stored, that the contents of memory are more abstract (but see Johnson, 1968, and Martin & Roberts, 1966, for a discussion of how surface structure might be implicated in sentence memory). Miller (1962) initiated the approach in which the contents of memory are equated with more remote linguistic entities; he hypothesized that sentences are stored as kernels (simple, active, affirmative, declarative sentences) with mental tags indicating the transformations necessary to transform the kernel back to its original form. More recently, these notions have been up-dated (Clark, 1969; Wanner, 1968) to hold that the abstract underlying structure (Chomsky, 1965) is what is stored in memory. (The latter two positions differ from Miller's in that Miller's kernel is transformationally derived from the underlying structure.) Going one step further, Bransford, Barclay, and Franks (1972) hypothesize that subjects store some representation of the semantic situation signaled by the sentence. Taking a different approach, Paivio (1971) proposes a hypothesis free of abstract syntactic forms; Paivio believes that an abstract sentence is stored as a string of words, while a concrete sentence may be stored in some imaginal-spatial form.

All of the above hypotheses have to do with the contents of memory, whose properties must be inferred through examination of overt

[1] This research was supported by Grant MH 19859 from the National Institutes of Mental Health.



Copyright © 1973 by Academic Press, Inc.
All rights of reproduction in any form reserved.

performance data. Let us consider one well-known inference about storage, drawn by Mehler (1963). Mehler claimed that sentences were stored in memory as a kernel plus transformational tags. To test this storage hypothesis, he presented a list of sentences, each having a different syntactic form, for subjects to memorize. Analysis of recall data indicated a general tendency to simplify the syntactic form of the original sentence. Mehler tacitly assumed that the recall process reproduced what happened to be in the memory store; if the tag was still stored, then no simplification should have occurred. However, recall is reconstructive, not merely a playback of the information stored in memory. Hence, Mehler's results may reflect the contents of memory, but they may also be due to "response biases." Clearly, it is desirable to know something about the operation of reconstruction if one wishes to make inferences about storage.

At least since Bartlett (1932), psychologists have been well aware of the active, reconstructive nature of recall, and many contemporary models of memory (cf. Rumelhart, Lindsay, & Norman, 1972) explicitly embrace the notion. However, little empirical insight into reconstructive processes has been gained in the interval since the appearance of Bartlett's book. We would like to advance the following simple notion about reconstruction at the single-sentence level: The syntactic constructions that characterize an individual's normal conversation will be reflected in his recall data, for normal speech production is itself a constructive activity whose end product depends, in part, upon the information stored in memory. If characteristic patterns can be identified, one can then use these, in conjunction with a storage hypothesis, to predict the form in which a sentence will be recalled.

### *Storage Hypothesis*

Gomulicki (1956) has speculated about an abstraction process by which items come to be represented in memory because of their relative contributions to a passage. He argued that the act of understanding and the act of abstracting for memory share the characteristic that parts of a discourse are (perhaps unwittingly) marked for importance. They differ, however, in that abstraction proceeds to eliminate enough of the less important elements so that the more important elements may be carried forward in memory.

Understanding a sentence consists of arriving at a semantic interpretation, a process involving, among other things, determination of the underlying or deep-structure relations of the sentence. It is assumed that some subset of the deep structure relations are abstracted for storage in memory. That is not to say that these relations are imprinted directly upon a passive mind. The information must be fit into existing cognitive structures (cf. Rumelhart *et al.*, 1972), and these structures influence the ultimate form in which the new information is stored. We agree with Bransford *et al.* (1972): " . . . sentences are not viewed as linguistic objects to be remembered. Instead they are viewed as information which subjects can use to construct semantic descriptions of situations. These constructed descriptions may contain more information than is represented in the linguistic inputs, and hence a purely linguistic analysis of the input sentences will not adequately characterize the information available to the listener [p. 194]." Nevertheless, the basic conceptual relations are preserved, and in the present experiment they cannot be distinguished from the deep-structure relations.

Surface-structure form, being only indirectly related to semantic interpretation, is in a sense discarded by the comprehension process, and is not customarily stored in memory. Recognition tests support this claim in cases where subjects are not given time to rehearse (Sachs, 1967), or do not expect a later test (Bransford & Franks, 1971); in such cases, subjects do poorly at recognizing the original surface-structure form of a sentence.

Abstraction might also involve the extraction of a theme, a central topic around which the remainder of a passage is organized. Certainly, it is reasonable to talk about abstracting the theme of a book, movie, or paragraph (Pompi & Lachman, 1967) and there is ample precedent for supposing that isolated sentences also have themes. The analysis of single sentences into a topic (generally the surface-structure subject) and a comment has a long linguistic history (Sapir, 1921). For example, Chomsky (1965, p. 221) has suggested that such an analysis might be more appropriate for characterizing surface-structure relations (withholding functional relations like "subject-of" for deep structure).

Clark and Card (1969) discuss several sources of evidence that seem to establish the psychological validity of theme, which, for isolated sentences, they equate with the surface-structure subject. Part of this evidence consists of the observation (Anderson, 1963; Clark, 1966; Coleman, 1965) that subject nouns are the best remembered word from active and passive transitive sentences. However, James (1972) argued that these studies probably confounded syntactic position with image value. The claim is based upon linguistic fact and a psychological fact. Linguistically, more transitive verbs require an animate subject than require an animate object. Psychologically, Clark (1965) found that asking people to compose active and passive transitive sentences resulted in more animate (and therefore high image value) subject than object nouns. Since animateness was not controlled in the above experiments, it is likely that the bias observed by Clark confounded the subject–object comparison.

James found that recall of subject and object nouns was largely a function of their relative image values. James concluded that if one equates theme with the word most frequently recalled (Clark & Card, 1969) a listener's theme is not determined by syntactic function; rather, selection is based upon context (situational or verbal), relative saliency (for example, image value) of the nouns in the sentence, and idiosyncratic preferences and interests. In the absence of relevant context, as in an experimental setting, relative saliency would be expected to exert a strong determining influence on the selection of a theme. If there is no context, and subject and object nouns have approximately equivalent saliency, the choice of theme will depend upon the individual subject's association to the words.

To this point, it has been assumed that a sentence's semantic interpretation is stored in memory, and that it includes a theme and basic logical relations; these elements, which can be likened to Bartlett's stimulus traces and key words, provide the raw material from which a recollection must be reconstructed. What does the reconstructive process fashion, given this input?

*Reconstruction Hypothesis*

Biases operating in normal discourse should also affect the surface-structure form of recalled sentences. One such bias has been discussed by Carroll (1958) and Clark (1965). Carroll noted that speakers tend to place the most salient aspect of an extralinguistic situation in the subject position of a sentence describing that situation. As mentioned above (Clark, 1965), when subjects are given sentence frames to complete, they are more likely to use an animate noun as subject than as object. Consequently, it is assumed that, when reconstructing a sentence from memory, recall is likely to be initiated with the most salient noun (the theme) that was abstracted from the original sentence.

Second, it has been commonly observed that ordinary conversation contains more active than passive sentences. It is reasonable that the same bias should be realized when constructing sentences from memory.

*Predictions*

In the research to be reported here, partial control over the selection of a theme (and hence reconstruction) was achieved by varying

image value (*I* value, Paivio, Yuille, & Madigan, 1968) of subject and object nouns. To this end, four types of sentences were used, defined by the orthogonal combinations of high- or low-*I* subject nouns with high- or low-*I* object nouns. The resulting sentences will be referred to as HH, HL, LH, and LL. (In all cases, coding refers to the surface-structure order of the nouns.) Active and passive forms of all four types of sentences were used in a free recall paradigm. Applying the assumptions about storage and reconstruction to this experimental situation, we would not expect the probability of incorrectly transforming an active sentence to its passive form, or vice versa, to be constant for the four sentence-types. Let us delineate the predictions for each sentence type. First, recall that we assume that a semantic representation containing a set of deep-structure relations and a theme is stored in memory. The same set of deep-structure relations is presumably stored for all sentence types, but the theme should vary with sentence-type. James' (1972) data indicate that the theme of HL sentences is likely to be the subject noun, of LH sentences, the object noun; subject and object noun would be chosen approximately equally often when *I* value is equated (HH and LL sentences).

For HL sentences (active and passive), the subject noun is generally taken as theme; placing the theme in subject position during reconstruction correctly reproduces the original sentence. For example, consider the sentence *The burglar was startled by the incident;* it is presumably stored (approximately) as *incident startled burglar*, with *burglar* as theme. A correct sentence should be reconstructed, since using *burglar* as surface-structure subject (first noun) will result in a passive sentence. Hence, the probability of an error in voice should be relatively low for HL sentences.

However, for LH sentences, placing the theme (which is likely to be the high-*I* object) in the position of surface-structure subject in recall results in a transformation to the opposite voice. For example, the sentence *The rumor was denied by the sultan* would be stored (approximately) as *sultan denied rumor* with *sultan* as theme. Lacking information in memory about surface-structure form, at recall *sultan* is placed in the subject position, producing *The sultan denied the rumor*. Hence, a high probability of an error in voice is predicted for LH sentences. The HH and LL sentences should be transformed with intermediate and approximately equal probability, since it is supposed that subject and object nouns are equally likely to be chosen as theme for both sentence-types.

The second characteristic of normal discourse identified above generates the prediction that the probability of transforming a passive sentence of any given type to active will be greater than the probability of transforming an active sentence of the same type to passive. Of course, this pattern of results was reported by Mehler (1963). The kernel-plus-tag explanation then favored by Mehler attributed the result to forgetting of the appropriate transformational tags that had been stored in memory. We believe that Mehler's result reflects a general property of reconstruction, rather than a property of memory storage. In support of this contention, an array of negative results for the general kernel-plus-tag notion have been reviewed by Fodor and Garrett (1966).

## Experiment I

### *Method*

*Materials.* Eight sets of 24 simple transitive sentences were constructed, as defined by the orthogonal combination of the variables of subject *I* (high, low), object *I* (high, low), and voice (active, passive). Each sentence consisted of three content words (subject, verb, object) and function words. The 384 nouns in these sentences were selected from Paivio, Yuille, and Madigan (1968); high-*I* nouns had ratings greater than 5.90; low-*I* nouns, less than 3.50. Mean frequency (Kučera & Francis, 1967) was approximately equated for all 16 sets of nouns (overall, 41.7).

The 192 sentences were divided into 24 test lists; each test list contained one of each of the eight sentence

types plus an initial dummy sentence to attenuate primacy effects. Sentence type was counterbalanced across lists so that all sentence types appeared equally often in all test positions (2–9). Half the subjects were presented with Form A of the sentences, half with Form B; the active sentences in Form A were passive in Form B, and passive sentences in Form A were active in Form B.

*Procedure.* Twenty undergraduates were paid $2.00 an hour to participate in groups of two to five in four 1-hour sessions. Ten subjects received Form A, ten, Form B. Lists were read with a 2-second interval separating the end of one sentence and the beginning of the next. Two seconds after the ninth sentence, subjects began shadowing a prose passage, and continued for 15 seconds. Shadowed passages were taken from Titchener (1917), and were judged dissimilar to the test sentences. (There were no instances where subjects recalled material from the shadowed passage.) A 3-minute written free recall period followed; instructions emphasized the importance of guessing and partial report. All subjects received a practice list and test lists 1–7 on Day 1, test lists 8–15 on Day 2, test lists 16–24 on Day 3, and test lists 1–9 on Day 4. (The experiment was originally planned for 3 days, but was extended when low error rates were observed. New sentences were not used, because it was increasingly difficult to form reasonable sentences subject to restrictions on *I* value.)

*Scoring.* For a sentence to be scored as correct, all content words had to be recorded in their proper order. Errors of tense, pluralization, or function words were ignored. If all content words were recalled, but in the wrong order (active as passive or vice versa) a complete transformational error was scored. If two content words were recalled (the third either omitted or replaced by an intrusion—whether synonymous or not), the sentence was scored as partially correct if the content words were in their original relation to one another; if not, a partial transformation was scored. For example, a partial transformation would be scored if, instead of the originally presented *The baron sold the automobile*, a subject recalled either *The automobile was sold*, or *The car was sold by the baron*, or similar combinations.

### *Results and Discussion*

Table 1 presents the mean number of sentences recalled correctly (all words in the correct order) as a function of sentence-type, out of 33 possible. Reliable effects were noted for subject *I*, object *I*, and their interaction, all $ps < .01$. The difference between number of actives and passives recalled (mean difference, 2.04) was not reliable, $F(1,18) = 2.97$ $p > .05$. The only other factor to attain significance was the three-way interaction of form, subject *I*, and object *I*, $F(1,18) = 15.84$, $p < .01$; the two-way interaction of subject and object *I* was confined to Form A.

The mean number of subjects, objects, and verbs recalled (based on total output) as a function of sentence type is also presented in Table 1. As usual, fewer verbs were recalled than either subjects or objects. Subject–object comparisons largely parallel those reported by James (1972). Significantly more subject than object nouns were recalled from HL active and HL passive sentences (two-tailed

TABLE 1

MEAN NUMBER OF SENTENCES, SUBJECTS, VERBS AND OBJECTS RECALLED AS A FUNCTION OF SENTENCE TYPE (EXPERIMENT I)

| Condition | Sentences | Subjects | Verbs | Objects |
|---|---|---|---|---|
| *Active sentences* | | | | |
| HH | 10.50 | 15.00 | 12.70 | 14.85 |
| HL | 6.85 | 14.10 | 9.00 | 10.15 |
| LH | 3.95 | 9.70 | 6.40 | 9.95 |
| LL | 2.10 | 6.35 | 3.60 | 4.55 |
| *Passive sentences* | | | | |
| HH | 9.90 | 17.15 | 13.10 | 15.95 |
| HL | 6.35 | 12.80 | 7.00 | 10.03 |
| LH | 3.60 | 9.50 | 8.35 | 11.30 |
| LL | 1.75 | 5.50 | 2.75 | 5.25 |

sign tests, $p < .01$. The subject–object comparison did not achieve significance (at the .20 level) for HH active or passive sentences, LL passive sentences, nor for LH active sentences. Significantly more object than subject nouns were recalled from the LH passive set, $p < .05$. For LL active sentences more subjects were recalled than objects, $p < .05$. All of these results replicate those reported by James (1972) with the exception of the LL active comparison, which was not significant in the previous study.

Again, equating theme with the word recalled most often from a set of sentences, it can be argued that the likely theme of a HL sentence is the subject, and of a LH sentence, the object. This claim is made for LH active sentences even though objects were not recalled significantly more often. The same result was reported by James (1972) and attributed to syntactic factors that could operate to attenuate the object's advantage. Subject and object appear to be selected approximately equally often as theme of HH sentences, and of LL passive sentences. The current data indicate that subjects may have a preference for subject themes from LL active sentences. This last result is difficult to interpret; it does not agree with previous results (James, 1972), and the comparison was not significant for LL passive sentences.

*Voice-error probabilities.* Five storage hypotheses will be considered, along with the predictions they appear to generate about the ordering of voice-error probabilities. These predictions are made on the basis of claims about sentence storage; it will be seen that accurate prediction cannot be attained on this basis alone, and that some assumptions about reconstruction are necessary.

The most simple-minded hypothesis would hold that a sentence is stored exactly as presented. Although intuition denies the plausibility of such a prediction perhaps it should not be cavalierly discounted in an experimental situation where subjects are asked to remember exactly what they hear, and are given some interval in which to concentrate on memorization. The hypothesis predicts a constant (and very low) probability of a voice error for all eight sentence types.

Miller (1962) and Mehler (1963) have proposed that a sentence is remembered as a kernel plus a tag. The transformational tags are assumed to be highly unstable, allowing the hypothesis to predict a greater probability of voice errors for passive than for active sentences. This hypothesis has no provision that would permit variations among the four types of active sentences or among the four types of passive sentences.

A third candidate would have a sentence's deep structure stored, with no indication of surface-structure form. When active and passive sentences are included in the list, production of the two forms at recall should largely be a matter of probability matching: Across the four sentence-types, approximately half the sentences of each type should be recalled in active voice, and half in passive voice. Like the surface-structure memory hypothesis, a constant (but higher) probability of committing a voice error is predicted for all eight sentence sets.

In the Bransford *et al.* (1972) terminology, the deep-structure hypothesis is referred to as the interpretive hypothesis of sentence storage. It differs from their constructive hypothesis, as pointed out in the introduction, but both hypotheses make the same predictions in the present context. The constructive hypothesis is also, of course, just concerned with the contents of memory, and predicts a generally high probability of transformation.

A fifth hypothesis can be taken from Paivio's writings (Begg & Paivio, 1969; Paivio, 1971). According to Paivio, concrete sentences (both nouns concrete) are stored as images, with abstract sentences (both nouns abstract) stored as a sequence of verbal units. Paivio (1971) has been quite specific on the latter point: "The meaning of abstract material, however, is tied to the wording itself, and to remember the meaning is to remember the

specific wording [p. 24]." In the present experiment, the Paivio hypothesis predicts asymmetries in the likelihood of voice errors. In particular, the LL sentences should rarely be transformed; if they are stored as sequences of verbal units, subjects should not often recall an active sentence in the passive voice or a passive sentence in the active voice. On the other hand, if concrete sentences are coded primarily in an imaginal-spatial manner that is essentially nonsequential, a large number of the HH sentences of both active and passive forms should be incorrectly transformed at the time of recall. Since HL and LH sentences are intermediate in concreteness, the probability of a voice error on these sentences would presumably be intermediate; also, nothing in Paivio's writings would seem sufficient to suppose these probabilities different. Neither is it clear that a higher probability of a voice error should be observed for active as opposed to passive sentences. Hence, these predictions contrast sharply with those stated in the introduction.

The primary datum for choosing among these five alternatives is the probability of incorrectly transforming an active sentence to passive, or vice versa. This probability was defined as the ratio of the number of sentences recalled in a transformed version, either complete or partial (as distinguished in Method section) over the total number of sentences recalled in complete or partial form. These probabilities (and the data from which they were calculated) are presented in Table 2.

There is no completely appropriate statistical analysis for these data because of the many cells with no entries (67 out of a possible 160). Furthermore, the overall probabilities are not independent, since all subjects contribute data to each total. In spite of the latter fact, and in order to gather some idea about the variability among sentence types, separate chi-square tests were performed on the active and passive sets. The expected number of transformations for a given sentence type was defined as the product of the overall probability of transforming an active or passive sentence (collapsing across sentence type) and the number of items recalled having at least two content words. For example, the overall probability of transforming an active sentence to passive was .093. There were 284 HH active sentences recalled, so the expected number of transformational errors

TABLE 2

TOTAL NUMBER OF CORRECT AND TRANSFORMED SENTENCES RECALLED (COMPLETELY OR PARTIALLY) AND PROBABILITY OF A TRANSFORMATION ERROR (EXPERIMENT I)

| Condition | Completely correct sentences | Partially correct sentences | Complete transformations | Partial transformations | Probability of a transformation |
|---|---|---|---|---|---|
| *Active sentences* | | | | | |
| HH | 210 | 59 | 10 | 5 | .053 |
| HL | 137 | 58 | 3 | 3 | .029 |
| LH | 79 | 47 | 26 | 15 | .245 |
| LL | 42 | 25 | 5 | 0 | .070 |
| *Passive sentences* | | | | | |
| HH | 198 | 68 | 21 | 35 | .174 |
| HL | 127 | 68 | 5 | 5 | .049 |
| LH | 72 | 44 | 31 | 27 | .333 |
| LL | 35 | 32 | 10 | 3 | .163 |

was taken to be 26.41. (Only 15 were observed.) Both active, $\chi^2(3) = 55.75$, and passive sentences, $\chi^2(3) = 43.67$, appear to differ with respect to the relative frequency of transformational errors. (Exact probability values cannot be given, since the distribution of these pseudo chi-squares is not known.)

Were we to make the simple tacit assumptions that have characterized past studies of sentence memory (i.e., that the data directly reflect the contents of memory), we would be tempted to reject all five of the storage hypotheses discussed. The surface-structure hypothesis predicted an equal probability of transformation across all eight sentence types, as did the deep-structure and constructive hypotheses. The kernel-plus-tag hypothesis predicted lower values for active than for passive sentences, but no differences within each set. The data are clearly inconsistent with these predictions. Paivio's reasoning led to an expectation of a greater probability of transformation for HH than for LL sentences, and equal, intermediate values for HL and LH sentences. Just the opposite was obtained. (Actually, it might be objected that Paivio's two-process hypothesis does not necessarily make the predictions ascribed to it. Since a concrete sentence can be stored either as a picture or a string of words, perhaps concrete sentences are stored both ways; then the picture is retrieved and the associated string of words is read out. Taking this option, the Paivio hypothesis would appear to predict the same transformation probability for all sentence types. One way or the other, the hypothesis is not supported by the present data.)

However, the results agree very well with the predictions made by the deep-structure and the constructive hypotheses, when supplemented by the reconstruction assumptions. For both active and passive sentences, the order predicted in the introduction was obtained; subjects were most likely to incorrectly transform a LH sentence, less and approximately equally likely to transform HH and LL sentences, and least likely to transform an HL sentence. This pattern is particularly clear for the passive sentences, where there are clear separations among the three groupings. For active sentences, LH sentences have a relatively high transformation probability, but few voice errors were made on the other three sentence types, although the ordering of the probabilities is as predicted. The data support the second prediction also: For all four sentence types, transformation probability was higher for passive than for active sentences.

## Experiment II

Experiment I partially determined choice of theme, and hence reconstruction, through manipulation of the *I* value of subject and object nouns. It would be desirable to demonstrate that these results were not wholly peculiar to *I* value. Normally, situational variables, in the broad sense, largely determine which part of a sentence will be taken as theme; one of these is the speaker's intonation and emphasis.

Experiment II held subject and object *I* value constant (only HH sentences were used) and attempted to determine the theme by varying the emphasis placed on subject and object nouns. In each list one-third of the sentences were preceded by the subject noun (S sentences), one-third by the object noun (O sentences), and the remainder were preceded by silence (U sentences). Instructions specified that a word preceding the sentence should be taken as the theme, and subjects were urged to concentrate on that word. This procedure was intended to affect the representation of the sentence in memory by influencing the choice of theme. It was expected that emphasizing a word would increase its probability of occurring as surface-surface subject. Biasing the subject as theme should result in a decrease in transformation probability relative to the control condition (no words emphasized). The greatest value should be noted for sentences whose objects were emphasized. The same pattern, $S < U < O$, should be observed for active and passive sentences, but the latter sentences should be more likely to undergo transformation.

### *Method*

*Materials.* Each sentence contained a high-*I* subject and object; mean *I* value (overall, 6.19) and mean frequency (overall, 44.3) were approximately equated for subject and object nouns in both active and passive sentences. Of the 96 sentences, 40 were taken from the

set of HH sentences used in Experiment I. Eight lists each contained two sentences from each of the six conditions of the experiment (active or passive voice; subject, object, or neither emphasized). Each sentence was always used in the same voice. Across subjects, each sentence was used equally often in each emphasis condition; hence, there were three forms of the basic list of 96 sentences. Serial position during presentation was counterbalanced across subjects; overall, each of the six sentence types occurred equally often in each of the 12 serial positions.

*Subjects.* Rutgers College underclassmen enrolled in Introductory Psychology were tested singly or in pairs. The experiment took one hour and subjects were offered their choice of $2.00 or credit toward fulfilling their course quota of required experimental hours. Of 36 subjects, 25 took the money.

*Procedure.* All sentences were read with normal intonation, 2 seconds of silence following each sentence. If subject or object noun was emphasized, the critical word was read 1 second before the sentence (*baron—The baron sold the automobile*; or *automobile—The baron sold the automobile*). Nothing was said preceding the unemphasized sentences. Hence, total presentation and study time was less for U sentences. The instructions were as follows:

> Sometimes I will say a word before the sentence. This is to indicate that I consider this word to be the theme or focus of the sentence; that is, linguists hold that all sentences can be analyzed into a topic—what the sentence is about—and a comment about the topic. For some of these sentences, I will indicate the topic which I had in mind when I wrote the sentence. Concentrating on this word while studying the sentence should make the sentence easier to remember. I will not say a word before some sentences. This is because I had no particular theme in mind for these sentences.

After the twelfth sentence of each list, subjects counted backward by threes from a three-digit number for 15 seconds; a 3-minute written recall period followed, during which guessing and partial report were encouraged.

### *Results and Discussion*

Table 3 presents the mean number of sentences recalled correctly per subject (all content words in the proper relations) for the six conditions, out of 16 possible. Analysis of variance revealed the only significant effect to be that of emphasis, $F(2,66) = 3.23$, $p < .05$. Fewer U sentences were recalled, probably because, as noted in Procedure section, longer study intervals accompanied the presentation of S and O sentences. Table 3 also includes the mean number of subjects, verbs, and objects recalled under each condition. Again, fewer verbs were recalled from U sentences, $F(2,66) = 7.32$, $p < .01$.

Analysis of object recall also revealed a significant effect of emphasis, $F(2,66) = 21.09$, $p < .01$; by Tukey's $a$ test, reliably more objects were recalled from O sentences, both $p$s $< .05$, and approximately the same number were recalled from S and U sentences, $p > .05$. Analysis of subject recall indicated an effect of emphasis, $F(2,66) = 15.62$, $p < .01$, and an interaction between emphasis and the form of the list, $F(4,66) = 3.66$, $p < .01$. In two forms,

TABLE 3

MEAN NUMBER OF SENTENCES, SUBJECTS, VERBS AND OBJECTS RECALLED AS A FUNCTION OF SENTENCE TYPE (EXPERIMENT II)

| Condition | Sentences | Subjects | Verbs | Objects |
|---|---|---|---|---|
| *Active sentences* | | | | |
| S | 3.28 | 7.03 | 4.53 | 5.00 |
| O | 2.94 | 5.53 | 4.53 | 6.92 |
| U | 2.89 | 5.81 | 3.72 | 4.81 |
| *Passive sentences* | | | | |
| S | 3.14 | 6.83 | 4.44 | 5.64 |
| O | 2.75 | 5.53 | 4.78 | 7.08 |
| U | 2.44 | 5.25 | 3.89 | 4.64 |

the ordering obtained was S > U > O; in the third form, there were more subjects recalled from O sentences than from U sentences. Overall, there was little difference between these latter two conditions.

More subject than object nouns were recalled from both active and passive S sentences, $ps < .01$ by two-tailed sign tests; conversely, more object than subject nouns were recalled from active and passive O sentences, both $ps < .01$. It would appear that the experimental operation of emphasis was effective in determining theme or salience. For U sentences, a greater number of subjects were recalled than objects; the difference was significant for active sentences, $p < .01$ by two-tailed sign test, but not for passive sentences, $p > .10$. Previous research (James, 1972, and Experiment I here) had revealed no reliable difference between number of subject and object nouns recalled from HH sentences, suggesting that the design of the present experiment influenced subjects to concentrate on the subject noun when no word was emphasized. If so, then U sentences were functionally equivalent to S sentences, and the data does not meet the requirements necessary for predicting a smaller transformation probability for S than for U sentences.

Table 4 presents the total number of transformations (complete and partial) observed for each condition, along with the conditional probabilities. Sentences were more likely to be transformed from passive to active than vice versa. This result is attributed to the bias which, in the absence of surface-structure information in memory, favors the construction of active rather than passive sentences.

As in Experiment I, separate chi-square analyses were performed on the active and passive sentences, although we did not have independent groups. Both analyses appear to allow rejection of the hypothesis that the three proportions are equivalent: for active sentences, $\chi^2(2) = 23.45$; for passive sentences, $\chi^2(2) = 9.07$. As predicted, emphasizing the object noun when presenting a sentence increased the likelihood that the object noun would be used as surface-structure subject in the reconstructed version of the sentence. Parallel effects were not obtained for the subject-emphasis condition; probability of transformation was not reduced (relative to the unemphasized condition) for either the active or passive forms. As indicated above, this failure is attributed to the functional equivalence of U and S sentences, under these experimental conditions. For the HH sentences used, transformation probability is appa-

TABLE 4

TOTAL NUMBER OF CORRECT AND TRANSFORMED SENTENCES RECALLED (COMPLETELY OR PARTIALLY) AND PROBABILITY OF A TRANSFORMATION ERROR (EXPERIMENT II)

| Condition | Completely correct sentences | Partially correct sentences | Complete transformations | Partial transformations | Probability of a transformation |
|---|---|---|---|---|---|
| *Active sentences* | | | | | |
| S | 118 | 82 | 13 | 8 | .095 |
| O | 106 | 64 | 26 | 29 | .244 |
| U | 104 | 66 | 11 | 7 | .086 |
| *Passive sentences* | | | | | |
| S | 113 | 63 | 26 | 20 | .211 |
| O | 99 | 52 | 44 | 35 | .338 |
| U | 88 | 60 | 17 | 20 | .213 |

rently underestimated for the unemphasized condition. Hence, we conclude that our experiment does not represent a fair test of the prediction that emphasizing the subject will reduce the probability of transforming an HH sentence.

## Conclusions

These experiments clearly indicate the role that reconstruction plays in sentence recall, and underscore the point that recall data cannot be unambiguously interpreted without taking these processes into account. (Of course, the same general caveat applies to recognition experiments.) None of the sentence-storage hypotheses can predict the results of Experiments I and II unless supplemented with some assumptions about reconstruction. With these assumptions, the constructive hypothesis (a sentence is stored as a semantic bundle having a theme and preserving logical relations among words) is consistent with the obtained results, as is the deep-structure hypothesis. Can the other storage hypotheses be successfully supplemented?

The kernel-plus-tag hypothesis would deny the operation of one of the reconstructive biases, preferring to use the notion of stored tags to predict the active-passive asymmetry in the probability of a transformation. If one accepts the principle of reconstruction, it would seem parsimonious (given the well-established preference for active sentences in spontaneous discourse) to interpret this result as reflecting an aspect of reconstruction rather than storage.

Paivio's storage hypothesis predicted differences for the four types of sentences in Experiment 1. However, the obtained order was the opposite of that predicted, and it is not clear that the predictions could be reversed by grafting on assumptions about reconstruction.

The surface-structure hypothesis can perhaps be modified to handle the results, if one assumes that the recalled form is determined by the reconstructive biases when the strength of the surface-structure trace falls below some minimal value. In itself, this is not unreasonable, but would require some elaboration as to the operation of trace erosion. It is rather curious that this hypothesis should be at all tenable, given the many reports (and intuitions) that argue against it. However, surface-structure form *was* recalled. If two or three words were remembered from a sentence, they were highly likely to be placed in their original relationship to one another: The probability correct varied from .662 to .971. Under the strong assumptions of the constructive hypothesis, these probabilities should not have been this high.

The reader might sense a contradiction between these data, which reveal generally good memory for an active–passive distinction, and previous work by Bransford and Franks (1971), Mehler (1963), and Sachs (1967), which report quite poor memory for such distinctions. However, the Bransford–Franks and Sachs studies tested memory in a recognition paradigm, and either no time was given for rehearsal of individual sentences, or subjects did not expect a later test. Neither of these conditions held in the present experiment, and it would appear that surface-structure information was stored, and well. It might be objected that neither of these conditions held in Mehler's (1963) recall experiment, which found a sizable number of incorrect transformations. But Mehler used eight syntactic forms, as opposed to the two used in the present experiments. Apparently, increasing the confusability of the sentences can increase the likelihood of a transformation.

Clearly, then, whether one finds good or poor memory for surface-structure distinctions depends in part upon task-induced strategies and demands: If subjects are asked to remember sentences verbatim, and given time to rehearse, they will do reasonably well at storing surface-structure form. (This is not to say that a surface-structure phrase marker is stored; perhaps the most likely explanation would be an increased integration of the sen-

tence at the phonological level.) Whatever the nature of the information, it must be presumed to dissipate more rapidly than the associated semantic information.

We do not, of course, believe that surface-structure form is customarily retained, just that it may be in specific experimental contexts. Ordinarily, a surface structure is reconstructed consistent with the semantic information remaining in memory and the form of the construction reflects the characteristic speech patterns of the speaker.

We have been assuming that subjects abstracted a theme from individual sentences, that the theme was determined by relative saliency (*I* value, Experiment I) and by emphasis established by the experimenter (Experiment II). The notion of theme need not be interposed between the differences in saliency and the data. One could simply assume that words are stored in a sentence's memory representation in a more or less vivid form depending upon their saliency. The results of Experiment I are then interpretable in terms of a bias towards initiating a sentence with the most salient word in memory. The results of Experiment II can be similarly interpreted under the reasonable assumption that repeating a word will increase its salience. However, since the notion of theme has widespread linguistic and psychological currency, it was decided to interpret the results in these terms. Future research will reveal whether the notion of theme is to be usefully applied to single sentences in an experimental context of this nature.

## References


ANDERSON, B. *The short term retention of active and passive sentences.* Unpublished doctoral dissertation, Johns Hopkins University, 1963.

BARTLETT, F. C. *Remembering.* Cambridge: Cambridge University Press, 1932.

BEGG, I., & PAIVIO, A. Concreteness and imagery in sentence meaning. *Journal of Verbal Learning and Verbal Behavior*, 1969, **8**, 821–827.

BRANSFORD, J. D., & FRANKS, J. J. The abstraction of linguistic ideas. *Cognitive Psychology*, 1971, **2**, 331–350.

BRANSFORD, J. D., BARCLAY, J. R., & FRANKS, J. J. Sentence memory: A constructive versus interpretive approach. *Cognitive Psychology*, 1972, **3**, 193–209.

CARROLL, J. B. Process and content in psycholinguistics. In R. A. Patton (Ed.), *Current trends in the description and analysis of behavior.* Pittsburgh: University of Pittsburgh Press, 1958.

CHOMSKY, N. *Aspects of the theory of syntax.* Cambridge, Mass.: M.I.T. Press, 1965.

CLARK, H. H. Some structural properties of simple active and passive sentences. *Journal of Verbal Learning and Verbal Behavior*, 1965, **4**, 365–370.

CLARK, H. H. The prediction of recall patterns in simple active sentences. *Journal of Verbal Learning and Verbal Behavior*, 1966, **5**, 99–106.

CLARK, H. H. Linguistic processes in deductive reasoning. *Psychological Review*, 1969, **76**, 387–404.

CLARK, H. H., & CARD, S. K. The role of semantics in remembering comparative sentences. *Journal of Experimental Psychology*, 1969, **82**, 545–553.

COLEMAN, E. B. Learning of prose written in four grammatical transformations. *Journal of Applied Psychology*, 1965, **49**, 332–341.

FILLENBAUM, S. On the use of memorial techniques to assess syntactic structures. *Psychological Bulletin*, 1970, **73**, 231–237.

FODOR, J., & GARRETT, M. Some reflections on competence and performance. In J. Lyons & R. J. Wales (Eds.), *Psycholinguistics Papers.* Chicago: Aldine, 1966.

GOMULICKI, R. Recall as an abstractive process. *Acta Psychologica*, 1956, **12**, 77–95.

JAMES, C. T. Theme and imagery in the recall of active and passive sentences. *Journal of Verbal Learning and Verbal Behavior*, 1972, **11**, 205–211.

JOHNSON, N. F. Sequential verbal behavior. In T. R. Dixon & D. L. Horton (Eds.), *Verbal Behavior and General Behavior Theory.* Englewood Cliffs, N. J. Prentice-Hall, 1968.

KUČERA, H., & FRANCIS, W. N. *Computional analysis of present-day American English.* Providence, Rhode Island: Brown University Press, 1967.

MARTIN, E., & ROBERTS, K. Grammatical factors in sentence retention. *Journal of Verbal Learning and Verbal Behavior*, 1966, **5**, 211–218.

MEHLER, J. Some effects of grammatical transformations on the recall of English Sentences. *Journal of Verbal Learning and Verbal Behavior*, 1963, **2**, 346–351.

MILLER, G. A. Some psychological studies of grammar. *American Psychologist*, 1962, **17**, 748–762.

PAIVIO, A. Imagery and language. In S. J. Segal (Ed.), *Imagery: Current cognitive approaches.* New York: Academic Press, 1971.

PAIVIO, A., YUILLE, J. & MADIGAN, S. Concreteness, imagery, and meaningfulness value of 925 nouns. *Journal of Experimental Psychology Monograph Supplement* 1968, **76**, (No. 1, Pt. 2).

POMPI, K. F., & LACHMAN, R. Surrogate processes in the short-term retention of connected discourse. *Journal of Experimental Psychology*, 1967, **75**, 143–148.

RUMELHART, D. E., LINDSAY, P. H., & NORMAN, D. A. A process model of long-term memory. In E. Tulving & W. Donaldson (Eds.), *Organization of memory*. New York: Academic Press, 1972.

SACHS, J. S. Recognition memory for syntactic and semantic aspects of connected discourse. *Perception & Psychophysics*, 1967, **2**, 437–442.

SAPIR, E. *Language*. New York: Harcourt-Brace, 1921.

TITCHENER, E. B. *A textbook of psychology*. New York: MacMillan, 1917.

WANNER, H. E. *On remembering, forgetting, and understanding sentences: A study of the deep structure hypothesis*. Unpublished doctoral dissertation, Harvard University, 1968.

(Received June 16, 1972)